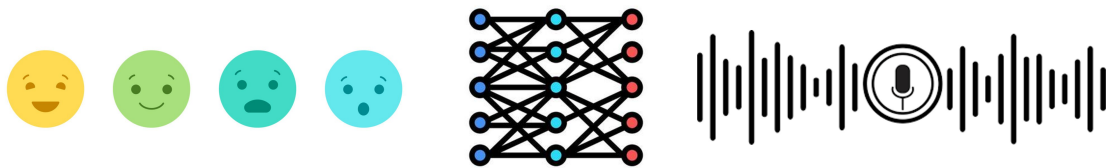


Artificial Intelligence for Digital Characters

Exercise 1: Affective Computing, Deep Learning & Speech Recognition

Hand-out Date: 11 March 2026



Goals

- Understanding the fundamentals and methodologies of affective computing.
- Understanding the key concepts of deep learning, such as neural networks, recurrent architectures, and attention mechanisms.
- Understanding the basics, techniques, and evaluation methods of speech recognition.
- Integrating speech recognition into a Unity project to enable an avatar to understand and respond to spoken commands.

Resources

The lecture slides, exercise slides, and Unity template are accessible via Moodle.

Tasks

1. Multiple Choice Questions

Please mark the correct answer for the multiple-choice questions below.

a) Multimodal sensing techniques are commonly employed in emotion recognition. Both facial expression analysis and blood pressure measurement are considered invasive sensing methods.

☐ True

☐ False

b) Action units can be used to detect emotions. Action units refer to the motion of the whole body.

☐ True ☐ False

c) ECG refers to recording electrical activity captured from the skin's surface.

☐ True ☐ False

d) The softmax activation function is commonly used in classification tasks. The output values of the softmax activation function are limited to the range $[-1, 1]$.

☐ True ☐ False

e) Recurrent Neural Networks (RNNs) process all time steps of a sequence simultaneously, making them highly parallelizable like Transformers.

☐ True ☐ False

f) Attention mechanism assigns different weights to different parts of the input, allowing the model to focus on important information.

☐ True ☐ False

g) Analog-to-digital conversion consists of two steps: sampling and quantization, where quantization converts real-valued numbers to positive numbers.

☐ True ☐ False

h) Pitch is the perceptual correlate of the fundamental frequency, which is the highest frequency component of a complex waveform.

☐ True ☐ False

i) A Mel filter bank is a set of overlapping rectangular filters used in signal processing, such as speech recognition and audio analysis.

☐ True ☐ False

2. Affective Computing

a) Please describe the James-Lange theory in your own words, and explain the difference to the Cannon-Bard and Schachter-Singer theories.

b) In the dimensional theory of affective computing, what are the meanings of valence, arousal, and dominance? How would you describe boredom using valence, arousal, and dominance?

c) Emotions can be detected from visual features like fidgeting. Given the following 3×3 matrices as the current frame f and the background image b :

$$f = \begin{bmatrix} 120 & 130 & 20 \\ 140 & 150 & 140 \\ 110 & 150 & 120 \end{bmatrix}, \quad b = \begin{bmatrix} 110 & 120 & 0 \\ 120 & 100 & 120 \\ 0 & 80 & 115 \end{bmatrix},$$

please compute the energy E and update the background image using the threshold $t = 5$ and the update ratio $\alpha = 0.2$.

d) Given the following heart rate sequences (in beats per minute) over a 5-minute period:

$$\text{seq}_1 = \{72, 75, 70, 73, 76\}, \quad \text{seq}_2 = \{74, 71, 76, 79, 80\}$$

Please calculate the standard deviation of the heart rate as an indicator of heart rate variability (HRV). How would you interpret a higher or lower value of HRV in terms of emotional states?

3. Deep Learning

a) Recurrent Neural Networks (RNNs) show promising performance in processing sequential data like text and audio. Please briefly explain how RNNs capture long-range information in sequential data.

b) A neural network unit with weights $\mathbf{w} \in \mathbb{R}^3$, bias $\mathbf{b} \in \mathbb{R}^1$, and activation function σ is defined as:

$$\mathbf{w} = \begin{bmatrix} 2.2 \\ -3 \\ 1.5 \end{bmatrix}, \quad \mathbf{b} = 3, \quad \sigma(x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases}.$$

Given an input $\mathbf{x} = [0 \ 5 \ 8]^T$, please compute the corresponding output of the neural network unit.

c) Positional encoding (PE) is widely used to capture the order information of tokens in Transformers. The PE formula is given by:

$$\text{PE}(t, 2i) = \sin\left(\frac{t}{10000^{2i/d}}\right), \quad \text{PE}(t, 2i+1) = \cos\left(\frac{t}{10000^{2i/d}}\right)$$

where t is the t -th position of the token, d is the dimension of the positional encoding, and i is the dimension index. Please compute the PE results for two 3-dimensional tokens (assume the indices of tokens and dimensions start from 1):

$$\text{token}_1 = \begin{bmatrix} 3.2 \\ -2 \\ 1.3 \end{bmatrix}, \quad \text{token}_2 = \begin{bmatrix} 1.2 \\ 3 \\ -1.5 \end{bmatrix}.$$

d) Given the query vector $q = [3 \ 1]^T$ and two pairs of {key, value} vectors $\{k_1, v_1\}$, $\{k_2, v_2\}$ with

$$k_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad v_1 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \quad \text{and} \quad k_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

please compute the attention weights (please use the scaled dot product as the similarity function) for each {key, value} pair and the output context vector.

4. Speech Recognition

a) Given the signal waveform shown in Fig. 1, please compute its amplitude, period, and frequency.

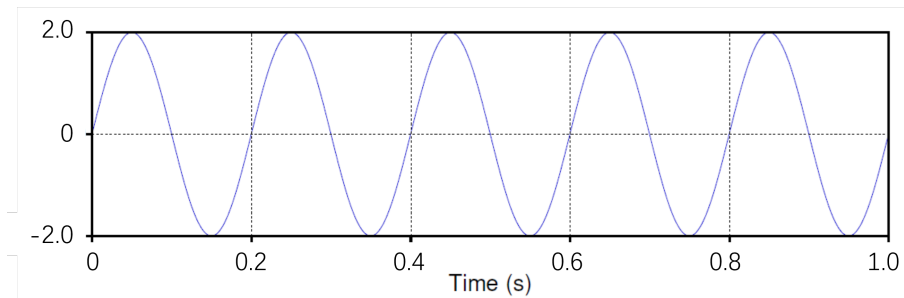


Figure 1: Example signal.

b) Given the lower and upper frequency bound as 100 Hz and 2000 Hz, respectively, please compute the start, peak, and end points in Hertz for four Mel filter banks. Hint: The Mel scale formula is as follows:

$$\text{Mels} = 1127 \ln \left(1 + \frac{f}{700} \right),$$

where f is the frequency in Hertz.

c) A speech recognizer often evaluates the Word Error Rate (WER) to evaluate the word string output. Given the two pairs of the reference string and the output string below:

Reference 1: could you get me a towel

Output 1: could you get me at all

Reference 2: hey tom it is mom will you come

Output 2: hey tom it is mom what do you call

Please compute the WER for each pair.

Project

Unity is a versatile game development platform widely used for creating interactive 2D and 3D experiences. In this project, you'll start with a Unity template featuring the Blender character Vincent and progressively integrate AI-driven functionalities and animations to develop your own digital character.

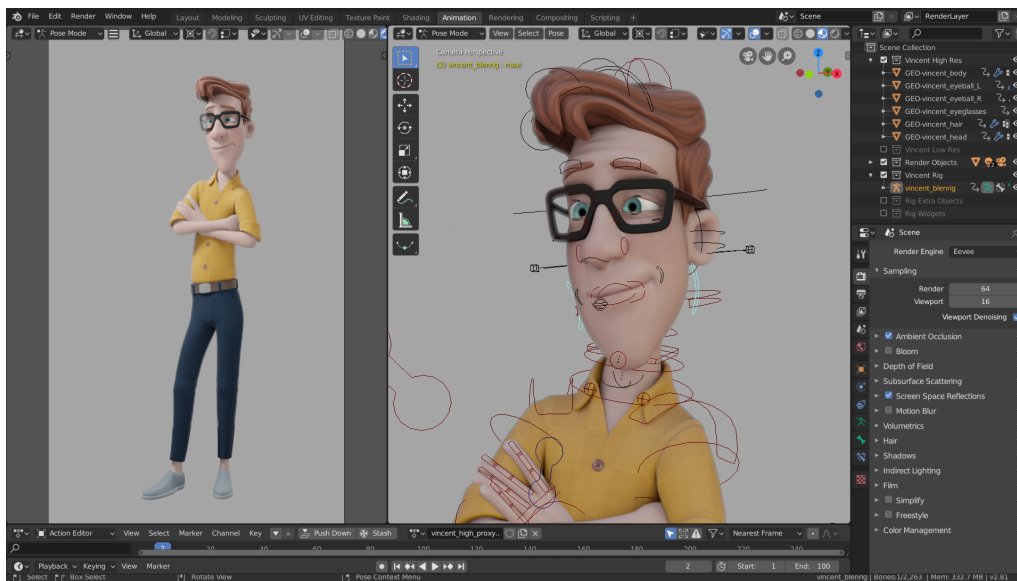


Figure 2: Digital character Vincent in Unity.

Project Stages:

1. **Speech Recognition Integration:** Implement speech-to-text capabilities using OpenAI's Whisper to transcribe spoken words into text. Display the transcriptions either in the console or on the screen.
2. **Large Language Model (LLM) Integration:** Connect a large language model, such as Llama or DeepSeek, to enable the character to generate text-based responses.
3. **Speech Synthesis Integration:** Incorporate speech synthesis technologies like FastSpeech 2 to provide a voice for the character.
4. **Character Animation Enhancement:** Enhance the character with animations, including facial expressions using tools like Audio2Face and body animations through platforms like Mixamo.
5. **Designing and Implementing Dialog Tree:** Create a dialog tree in Twine that links user inputs to narrative paths. Implement it in Rasa by defining intents, slots, and stories to ensure smooth transitions.

Stage 1: Integrating Speech Recognition

In this initial stage, you will familiarize yourself with Unity and enable Vincent to transcribe your speech. The process involves setting up a Python Flask server alongside OpenAI's Whisper for speech recognition. Flask serves as a lightweight web framework to handle communication between Unity and the speech recognition backend, allowing you to send audio data from Unity to the server for processing and receive the transcribed text in return.

1. Install Unity

- Download and install the Unity Hub from the [Unity Download page](#).
- Use the Unity Hub to install the latest version of the [Unity Editor](#).

2. Familiarize Yourself with Unity

- Learn the basics of using Unity. This [introductory video](#) and the [Unity Essentials](#) pathway might be helpful.
- Load the project template provided on Moodle into Unity to get acquainted with the project structure and initial setup.

3. Install Python and Whisper

- Download the latest version of Python from the [official Python website](#).
- Install the [Whisper API](#) model using the `whisper` package and other necessary dependencies.

4. Set Up the Python Flask Server

- Install [Flask](#) in your Python environment.
- Develop a basic Flask application capable of handling POST requests for audio files. An introductory video on using Python Flask with Unity can be found [here](#). You can find a template of a Flask application on Moodle.

5. Configure Flask to Handle Speech Input

- Create an endpoint in Flask to accept audio files (e.g., `@app.route("/recognize", methods=["POST"])`).
- Process the incoming audio file, convert it to a format compatible with Whisper, transcribe it, and return the transcribed text. You can test Whisper's functionality with audio files (e.g., `.wav` files).

6. Create Speech Recording in Unity

- Utilize Unity's microphone functionality to record audio (i.e., speech) from the user. You can use the `Microphone` class of Unity.

- Save the recorded audio as a file (e.g., `.wav`) that can be sent to the Flask server.

7. Set Up Unity to Communicate with Flask

- Use Unity's `UnityWebRequest` to send the recorded audio file as a POST request to the Flask API.
- Ensure the audio is sent in the correct format.

8. Display Transcribed Text in Unity

- Receive the transcribed text in Unity.
- Display the recognized speech as text in the console or on the screen.

9. Test the Integration

- Test the entire process, including recording, sending, and receiving data.
- Debug and ensure everything functions smoothly.

ChatGPT is a valuable resource to help you understand how Unity and Python Flask work. Should you have any questions, feel free to post them on the Moodle course forum. Enjoy working with Vincent!